

DATA ANALYST PORTFOLIO

AI Evaluation of Product Category Assignment and Factual Grounding Using Amazon ASIN Taxonomy

Prepared by:

Patrick Adu Osei

IBM Data Analyst Professional Certificate Holder

Tools & Technologies:

Python (Pandas, Numpy) | Matplotlib/Seaborn| Jupyter Notebook | GitHub

Date: June 2026

DATA ANALYST PORTFOLIO

1. Summary

This project evaluates AI-generated outputs for product category assignment using the Amazon ASIN-category dataset. The evaluation focuses on taxonomy correctness and factual grounding. The results from the visual analysis show that incorrect predictions significantly outnumber correct ones, indicating weak overall category assignment performance and poor factual grounding against the reference taxonomy.

Misclassification is concentrated in a small number of high-volume categories such as Clothing Shoes and Jewelry, Books, and Home and Kitchen. Additionally, the dominant failure type is unsupported claims, meaning that the system often assigns categories not supported by the ground-truth dataset rather than simply confusing similar categories.

2. Business/Analytical Question

The business and analytical questions the evaluation addressed are:

- Do AI outputs assign the correct product category?
- Which product categories are most prone to misclassification?
- How often do outputs contradict the product taxonomy?
- Which product contexts create factual grounding errors?

3. Dataset

The Amazon ASIN Category Dataset contains over 35 million product identifiers mapped to category labels. It provides a reliable ground-truth reference for validating AI-generated taxonomy assignments and assessing factual grounding performance.

4. Tools Used

- Python (pandas, numpy), Matplotlib, Seaborn, and Jupyter Notebook were used for data loading, evaluation, scoring, and visual analysis.

DATA ANALYST PORTFOLIO

5. Evaluation Process (What I Did)

1. A 5,000-representative sample of the dataset was selected to ensure computational efficiency.
2. AI-predicted categories were compared against true taxonomy labels for each ASIN.
3. Each prediction was scored across taxonomy correctness, factual grounding, context relevance, and unsupported claims.
4. Failure types were classified into structured categories, including category mismatch and unsupported claim.
5. Aggregated metrics and visualizations were generated to identify patterns and error concentrations.

6. Key Visualizations

1. Category Prediction Accuracy: Incorrect predictions far exceed correct predictions, indicating weak overall taxonomy accuracy and poor factual grounding.
2. Most Misclassified Categories: Errors are concentrated in a small group of categories, particularly Clothing Shoes and Jewelry, Books, and Home and Kitchen, suggesting structural classification difficulty in these domains.
3. Error Taxonomy Distribution: Unsupported claims dominate the error profile, significantly exceeding category mismatch, indicating that most errors are not grounded in the dataset.

7. Key Insights

1. Overall category assignment accuracy is weak, as incorrect predictions significantly exceed correct ones. This indicates poor alignment between AI outputs and the reference taxonomy.
2. Unsupported claims are the dominant failure type, showing that the system frequently generates categories without grounding in the dataset.
3. Error distribution is highly concentrated in a few categories, meaning misclassification risk is domain-specific rather than uniform.
4. The lack of contextual product information beyond ASIN contributes to poor classification performance.

DATA ANALYST PORTFOLIO

8. Business Implications and Recommendations (Why This Project Matters)

Incorrect category assignment introduces significant business risk by reducing search accuracy, weakening recommendation systems, and lowering product discoverability. The dominance of unsupported claims further introduces trust and reliability risks.

Recommendations:

- Prioritize factual grounding validation before optimization
- Incorporate product metadata such as titles and descriptions
- Focus improvement efforts on high-error categories
- Implement taxonomy validation rules and hierarchy checks

9. Challenges and How to Solve Them

The limited dataset context restricts accurate categorization. This can be addressed by incorporating richer product information. Overlapping category structures also introduce ambiguity, which can be mitigated using hierarchical taxonomy validation and multi-label classification.

10. What I would Improve Next

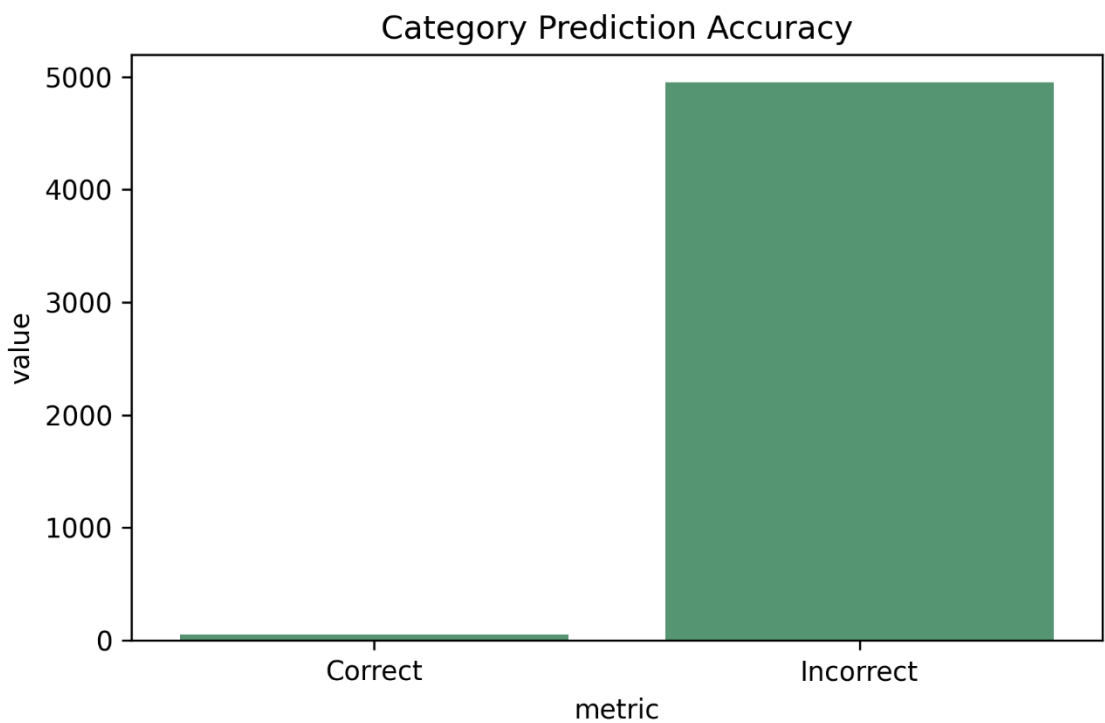
- Future improvements include integrating product metadata,
- Enabling multi-label classification,
- evaluating against human-labelled data, and
- refining error taxonomy to improve the capture of factual grounding failures.

11. Files and Links

- Notebook: ASIN category evaluation notebook (.ipynb)
Outputs: asin_category_accuracy.png, asin_misclassification.png,
asin_error_distribution.png

DATA ANALYST PORTFOLIO

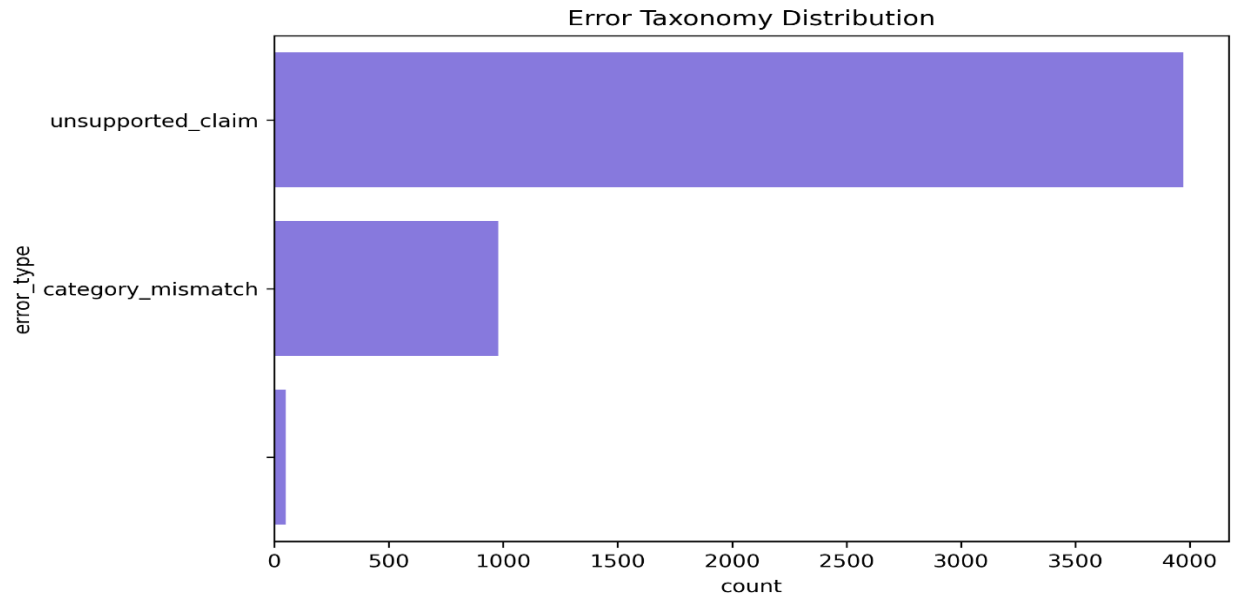
Chart 1: Category Prediction Accuracy



Insight: Incorrect category predictions significantly exceed correct ones, indicating **low taxonomy accuracy and weak factual grounding**. This suggests the AI output frequently fails to align with the reference product classification system.

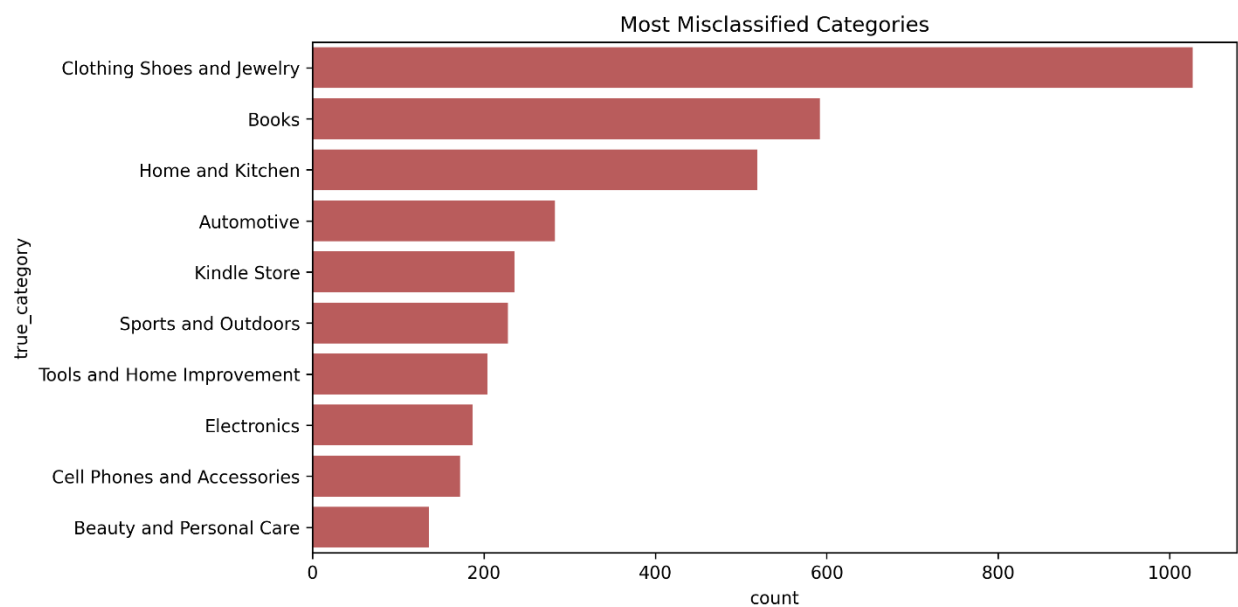
Chart 2: Error Taxonomy Distribution

DATA ANALYST PORTFOLIO



Insight: Unsupported claims dominate the error distribution, far exceeding category mismatches. This indicates that most errors arise from assigning categories that are **not grounded in the dataset**, rather than simple confusion between similar categories.

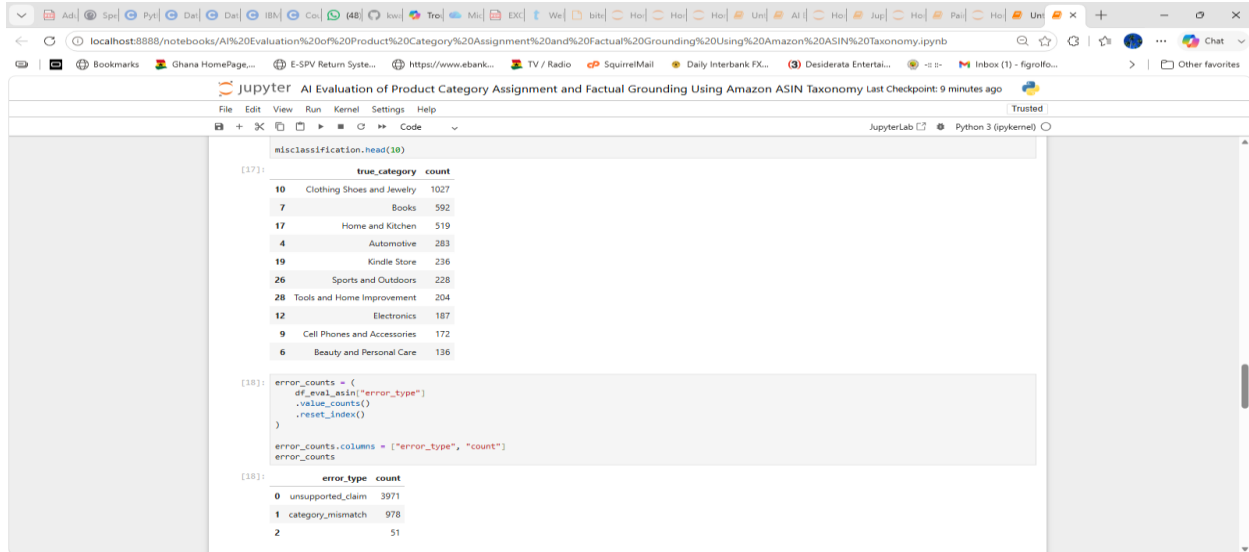
Chart 3: Most Misclassified categories



DATA ANALYST PORTFOLIO

Insight: Misclassification is heavily concentrated in a small set of categories, particularly **Clothing Shoes and Jewelry, Books, and Home and Kitchen**, indicating that broad and semantically overlapping product groups are more prone to classification errors.

Chart 4: Misclassification Categories



Insight: The underlying data confirms the visual patterns, showing high counts of unsupported claims and concentrated misclassification in specific categories, providing strong quantitative evidence of **systematic grounding failures and category-specific weaknesses**.

Overall visual takeaway:

Across all charts, the results show that the primary weakness is **poor factual grounding rather than correctness within known categories**, with errors concentrated in specific domains and dominated by unsupported category assignments.

Examples of Category Misclassification and Grounding Error

DATA ANALYST PORTFOLIO

ASIN	True Category	AI Predicted Category	Error Type
B07R3DYM6	Home and Kitchen	Electronics	Unsupported Claim

Insight:

In this example, the AI assigns the product to Electronics, while the true category is Home and Kitchen. This is classified as an unsupported claim because the predicted category is not grounded in the reference taxonomy and does not align with the actual product category.

ASIN	True Category	AI Predicted Category	Error Type
B09X1MRDN6	Clothing Shoes and Jewelry	Sports and Outdoors	Category Mismatch

Insight:

This example represents a category mismatch, where the predicted category is incorrect but still belongs to a plausible product grouping. Unlike unsupported claims, this error reflects confusion between related domains.